Solution

Problems: Impulse Response Properties and Convolution

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Impulse Response and LTI System Properties

1. A discrete-time system has impulse response $h[n] = (-1)^n u[n-3]$.

a) Is this system memoryless? requires $h[n] = c\,\delta[n]$

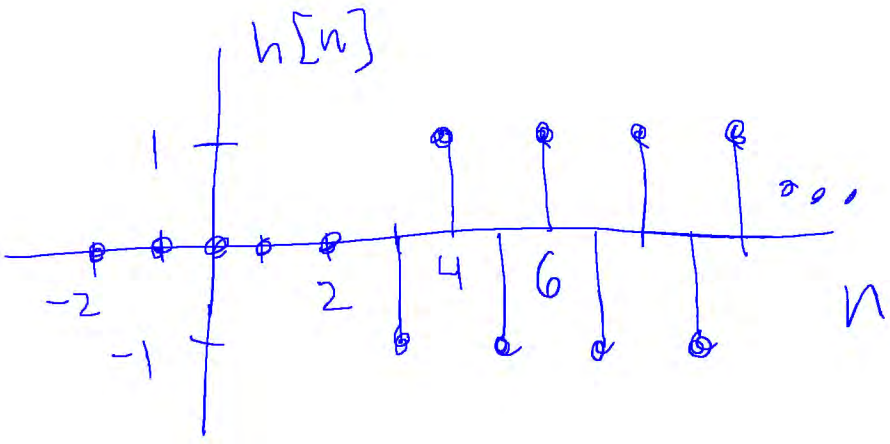
no, has memory

b) Is this system BIBO stable? requires $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$

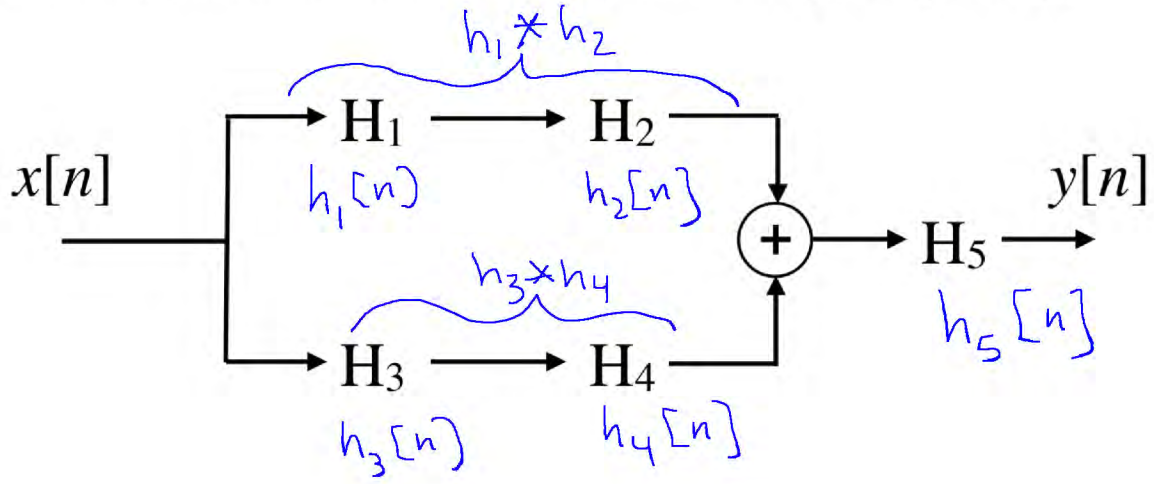
$\Rightarrow$ evaluate $\sum_{n=3}^{\infty} |(-1)^n| = \sum_{n=3}^{\infty} 1 \rightarrow \infty$ not stable

c) Is this system causal? requires $h[n] = 0$ for $n < 0$

yes, causal



2. In the system shown below H_i is an LTI system with impulse response $h_i[n[$.



Which of the following represents the impulse response $h[n]$ for the equivalent system relating $y[n]$ to $x[n]$? Select all that apply.

i) can't be determined from information given

ii) $h[n] = (h_1[n] * h_2[n] + h_3[n] * h_4[n]) * h_5[n]$

iii) $h[n] = h_1[n] * h_2[n] * h_5[n] + h_3[n] * h_4[n] * h_5[n]$

iv) $h[n] = h_5[n] * (h_3[n] * h_4[n] + h_1[n] * h_2[n])$

v) $h[n] = h_1[n] * h_5[n] * h_2[n] + h_5[n] * h_4[n] * h_3[n]$

vi) $h[n] = (h_1[n] * h_5[n] + h_5[n] * h_4[n]) * (h_2[n] + h_3[n])$

vii) $h[n] = (h_1[n] + h_2[n]) * (h_3[n] + h_4[n]) + h_5[n]$

associative

$h[n] = (h_1[n] * h_2[n] + h_3[n] * h_4[n]) * h_5[n]$

commutative

$= h_1[n] * h_2[n] * h_5[n] + h_3[n] * h_4[n] * h_5[n]$

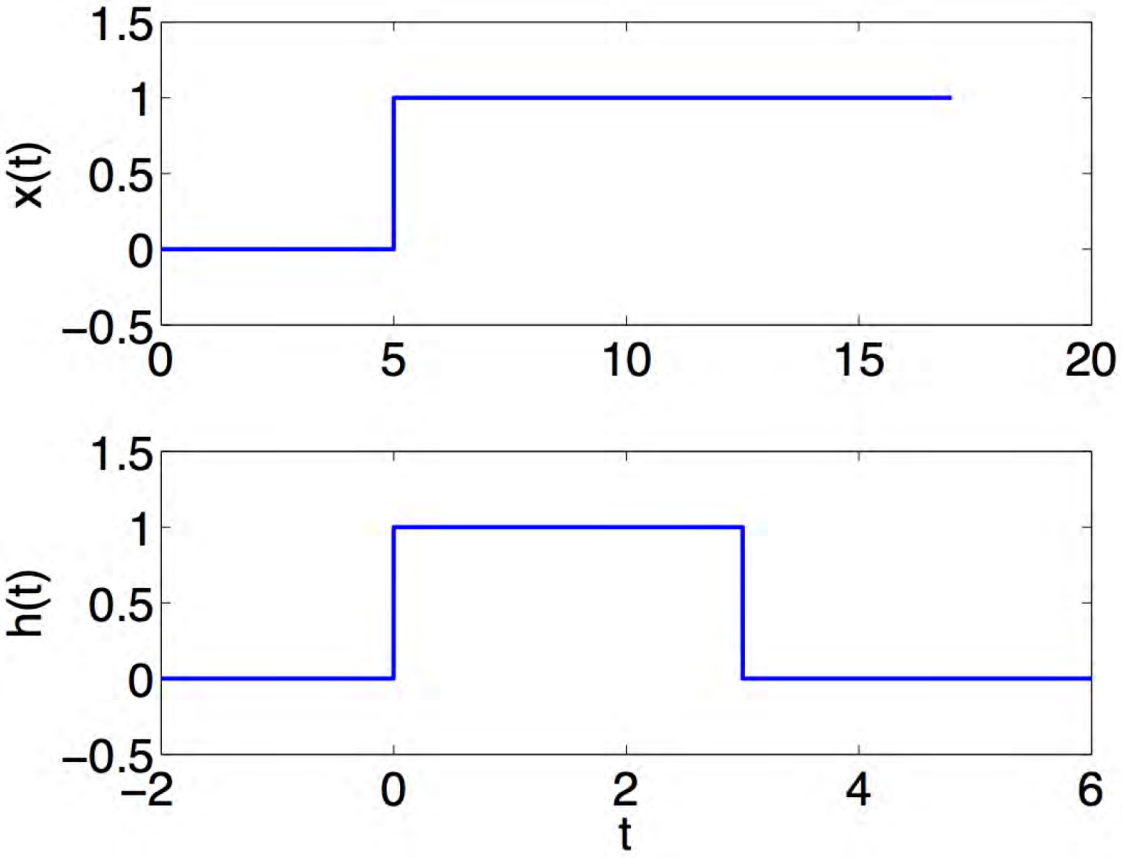
$= h_5[n] * (h_3[n] * h_4[n] + h_1[n] * h_2[n])$

$= h_1[n] * h_5[n] * h_2[n] + h_5[n] * h_4[n] * h_3[n]$

commutative

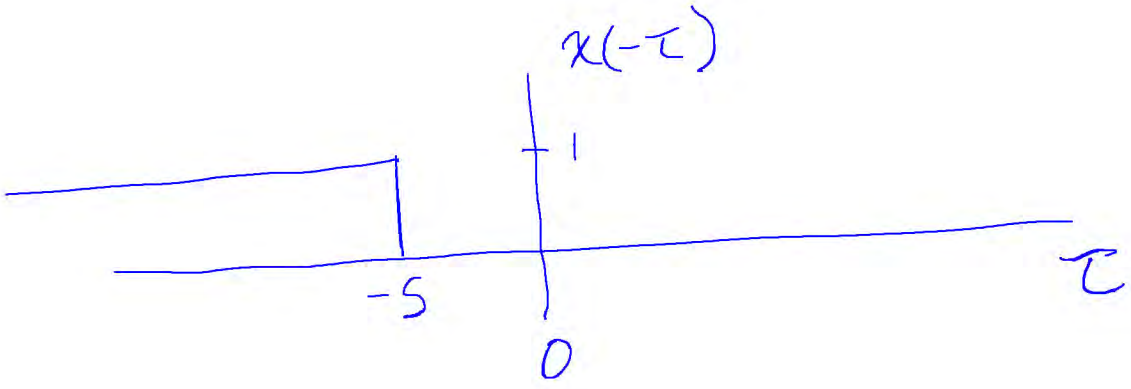
Convolution

3. In this problem we consider the convolution of continuous-time signals $x(t)$ and $h(t)$ where $x(t) = u(t-5)$ and $h(t) = u(t) - u(t-3)$ as shown below.

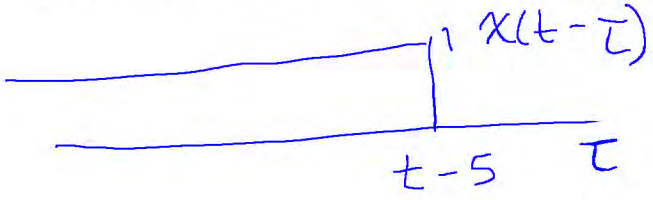


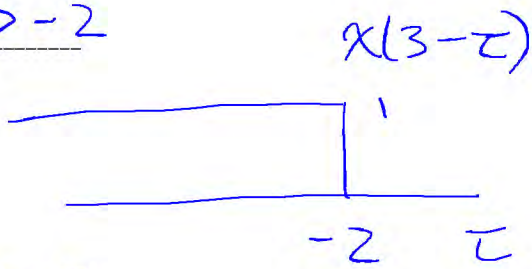
Complete the following statements:

a) Consider $x(-\tau)$. $x(-\tau) = 0$ for τ ____≥ -5

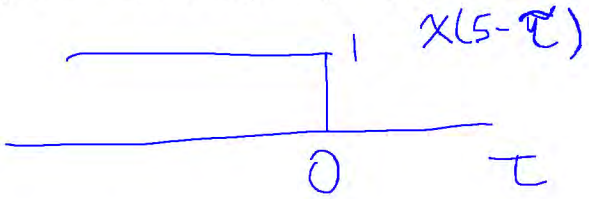


b) Consider $x(t-\tau)$. If $t=3$, then $x(3-\tau)=0$ for τ ≥ -2

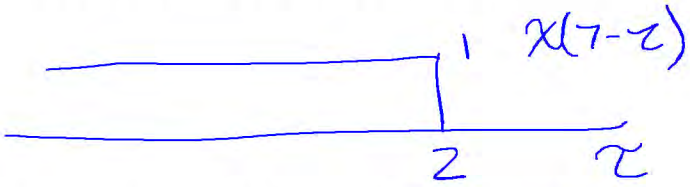




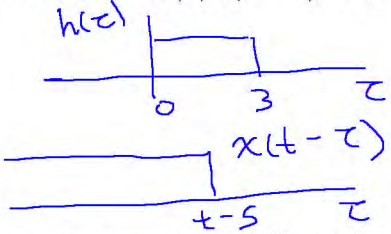
c) Consider $x(t-\tau)$. If $t=5$, then $x(5-\tau)=0$ for τ ≥ 0



d) Consider $x(t-\tau)$. If $t=7$, then $x(7-\tau)=0$ for τ _____



e) Now consider $h(\tau)x(t-\tau)$. For $h(\tau)x(t-\tau)\neq 0$ for some τ, we must have t > 5



require $t-5>0 \Rightarrow t>5$

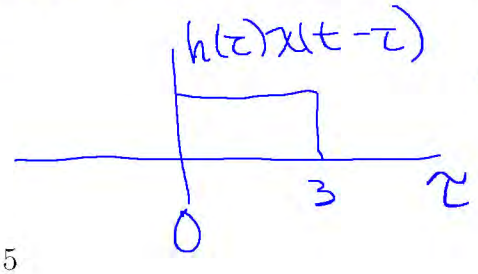
f) Now consider $y(t)=\int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$ and complete the statement $y(t)=0$ for t _____

$h(\tau)x(t-\tau)=0$ for $t<5 \Rightarrow y(t)=0$ for $t<5$

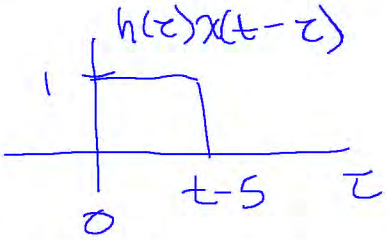
g) As t increases, $y(t)$ is non zero and constant if t is > 8

$y(t)$ constant $\Rightarrow$ area under $h(\tau)x(t-\tau)$ constant

for $t-5>3$ or $t>8$



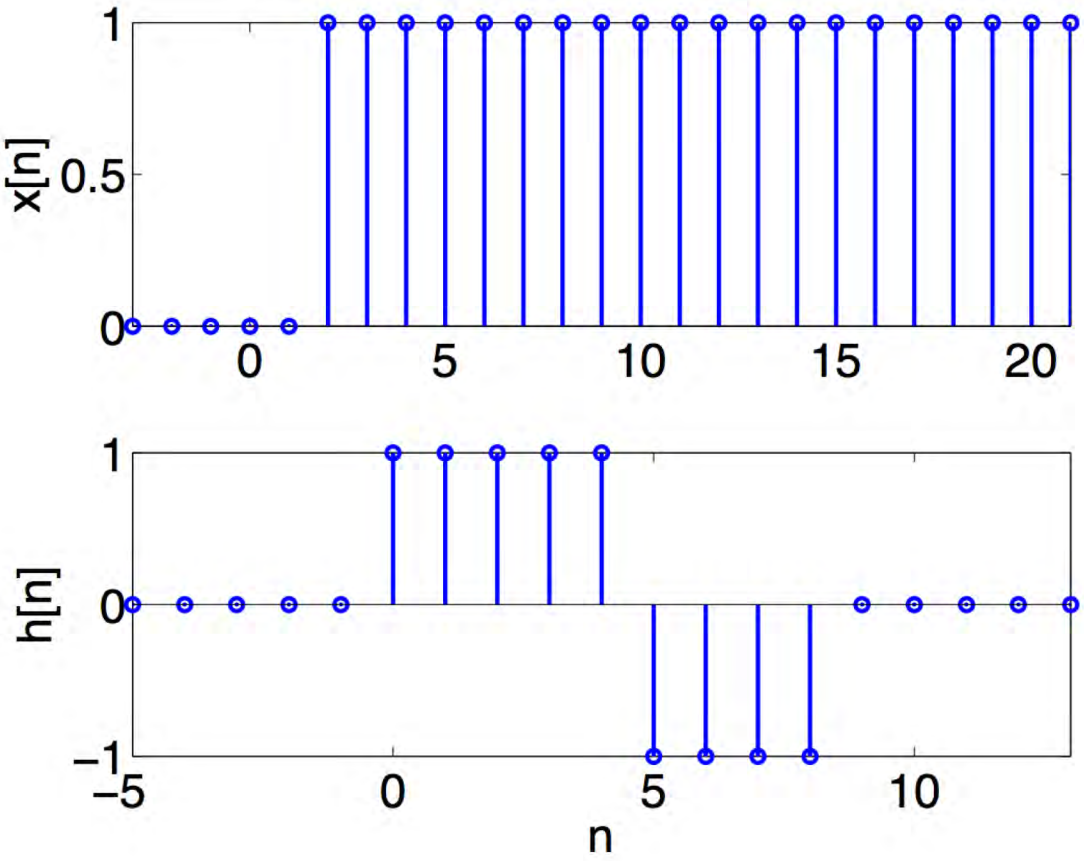
h) As t increases, $y(t)$ (increases, decreases) provided t is greater than 5 and less than 8



$5 < t < 8$

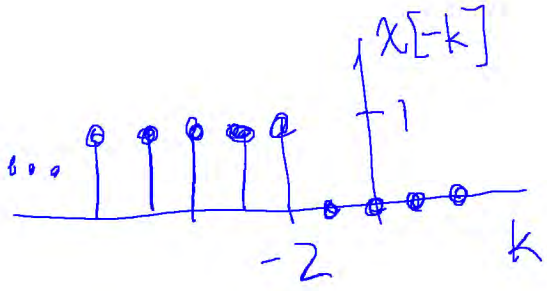
area increases as t increases

4. In this problem we consider the convolution of discrete-time signals $h[n] = u[n] - 2u[n-5] + u[n-9]$ and $x[n] = u[n-2]$ as shown below.

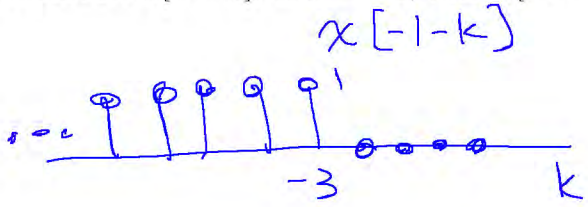


Complete the following statements:

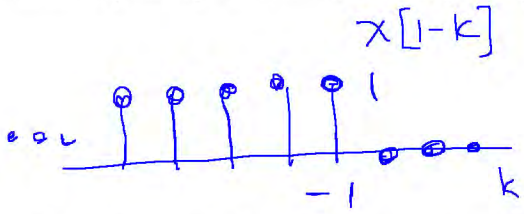
a) Consider $x[-k]$. $x[-k] = 0$ for k > -2



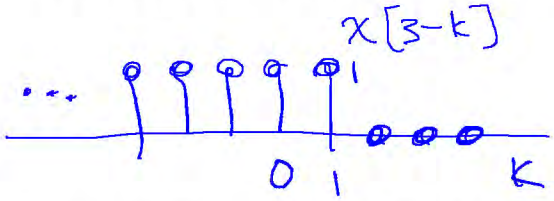
b) Consider $x[n-k]$. If $n=-1$, then $x[-1-k]=0$ for k ≥ -3



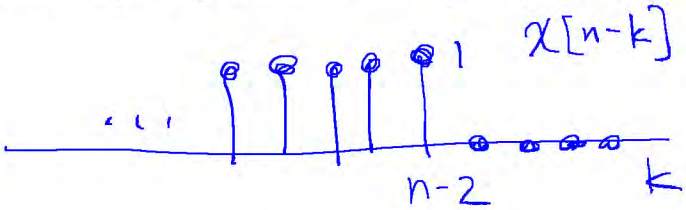
c) Consider $x[n-k]$. If $n=1$, then $x[1-k]=0$ for k ≥ -1



d) Consider $x[n-k]$. If $n=3$, then $x[3-k]=0$ for k ≥ 1

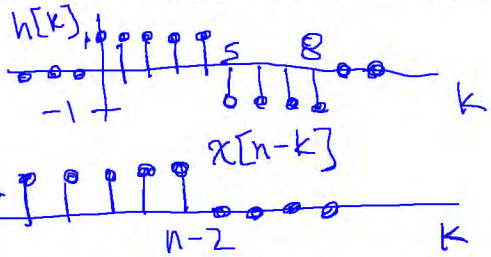


e) Consider $x[n-k]$. If $n=7$, then $x[7-k]=0$ for k ≥ 5



$x[n-k]=0$ for $k > n-2$

f) Now consider $x[n-k]h[k]$. The condition $x[n-k]h[k] \neq 0$ for some k, requires n be ≥ 2



require $n-2 \geq 0$

$n \geq 2$

g) Now consider $y[n]=\sum_{k=-\infty}^{\infty} x[n-k]h[k]$ and complete the statement $y[n]=0$ for n < 2

$y[n]$ is zero if $x[n-k]h[k]=0$ for all k

$\Rightarrow n-2<0$ or $n<2$

h) As n increases, $y[n]$ remains constant and non zero provided n is ≥ 10

$x[n-k]h[k]$ not changing with n, non zero area. For $n \geq 10$ $y[n] = 5 - 4 = 1$

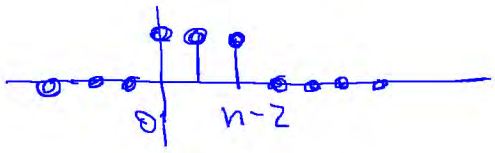
$n \geq 10$

$(n-2 \geq 8)$

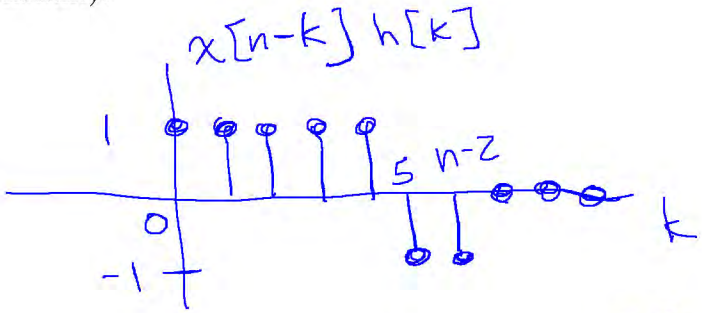


i) Consider the interval $2 \leq n \leq 6$. As n increases, $y[n]$ (increases, decreases, is constant).

$x[n-k]h[k]$ $y[n] = \sum_{k=0}^{n-2} 1$ for $2 \leq n \leq 6$

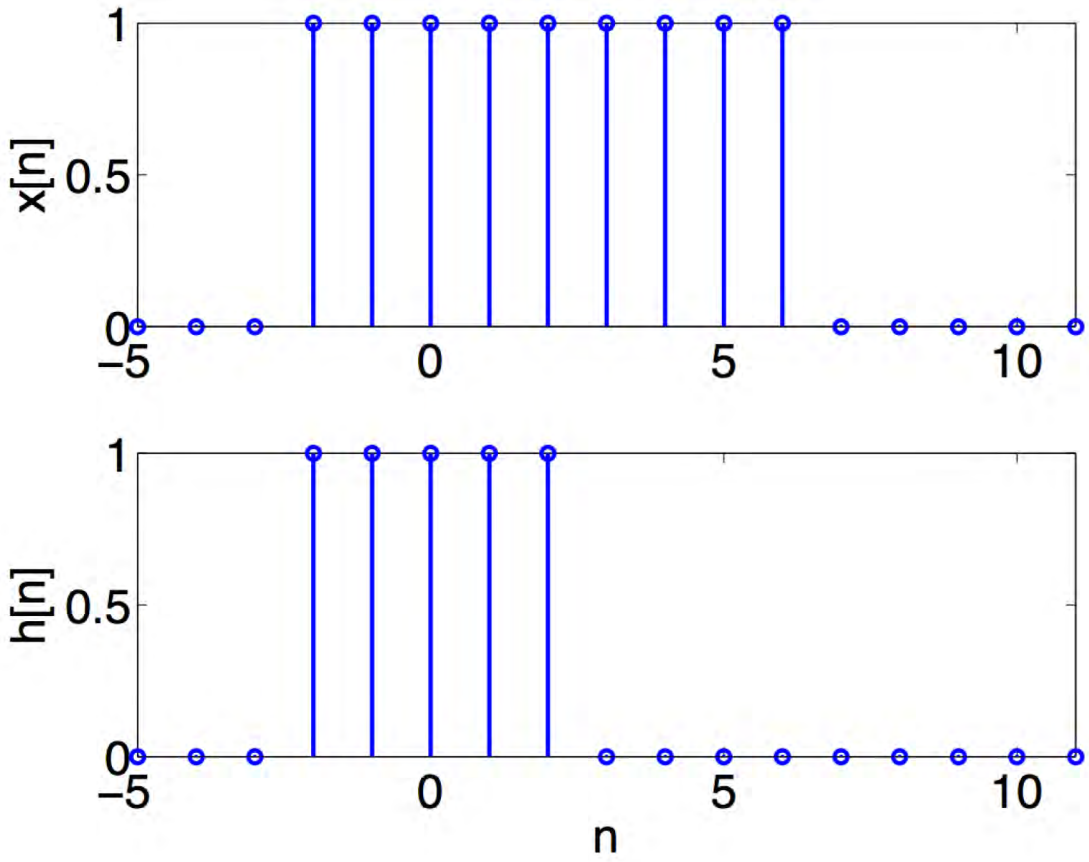


j) Consider the interval $6 \leq n \leq 9$. As n increases, $y[n]$ (increases, decreases, is constant).



$$y[n] = \sum_{k=0}^{4} 1 + \sum_{k=5}^{n-2} (-1) = 5 - (n-6)$$

5. Consider the convolution of discrete-time signals $x[n] = u[n+2] - u[n-7]$ and $h[n] = u[n+2] - u[n-3]$ as shown below.



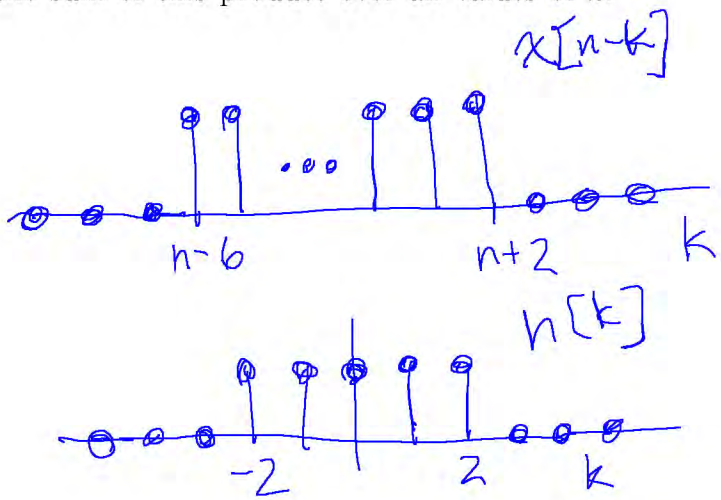
Complete the following statements. Hint - consider the product $x[n-k]h[k]$ for different values of n, and remember convolution is the sum of this product over all values of k: $y[n] = \sum_{k=-\infty}^{\infty} x[n-k]h[k]$.

a) For n less than -4 $y[n] = 0$.

$x[n-k]h[k] = 0$

for $n+2 < -2$

$n < -4$



b) As n increases beyond this point, $y[n]$ (increases, decreases, is constant) until $n =$ 4

$n \geq -4$



$y[n] = \sum_{k=-2}^{n+3} 1$ until $n+2 = 2$

$n = 4$

c) As n continues to increase, $y[n]$ (increases, decreases, is constant) until $n =$ 8

$n > 4$

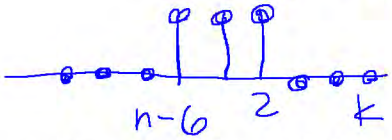


$y[n] = 5$ until $n-6 > 2$

$n > 8$

d) As n continues to increase, $y[n]$ (increases, decreases, is constant) until $n =$ 9

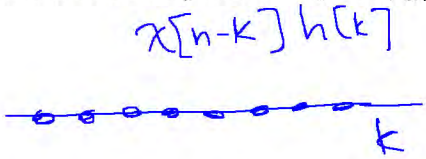
$n > 8$



$y[n] = \sum_{k=n-6}^{2} 1$ decreases as n increases, goes to 0 when $n-6 = 3$

e) As n continues to increase beyond this point, $y[n]$ (increases, decreases, is constant).

$n \geq 9$



$y[n] = 0$

f) For $n > 100$, the value of $y[n]$ is 0

6. Suppose an LTI system has impulse response $h(t) = e^{-t^2}\cos(\pi t)u(t+3)$. What is the output of this system $y(t)$ if the input $x(t) = 2\delta(t-2)$? Hint: use linearity, time invariance, and the meaning of the impulse response.

a) $y(t) = 2e^{-t^2}\cos(\pi t)u(t+1)$

b) $y(t) = 2e^{-(t-2)^2}\cos(\pi(t-2))u(t+1)$

c) $y(t) = 2e^{-(t-2)^2}\cos(\pi(t-2))u(t+3)$

d) $y(t) = 2e^{-(t+2)^2}\cos(\pi(t+2))u(t+5)$

e) $y(t) = 2e^{-(t+2)^2}\cos(\pi(t+2))u(t+3)$

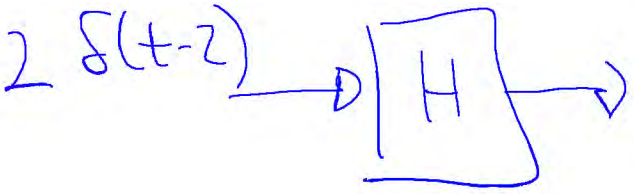
f) $y(t) = 2e^{-t^2}\cos(\pi t)u(t+5)$

g) $y(t) = 2e^{-t^2}\cos(\pi t)u(t+3)$



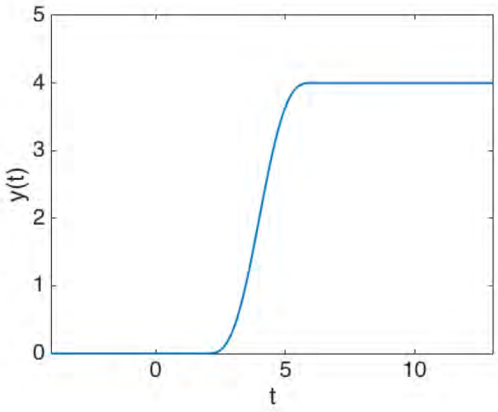


by time invariance

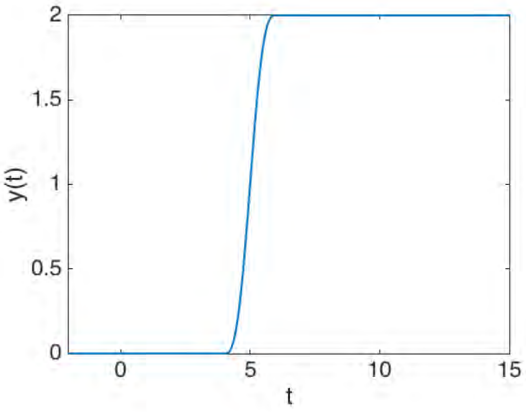


by linearity

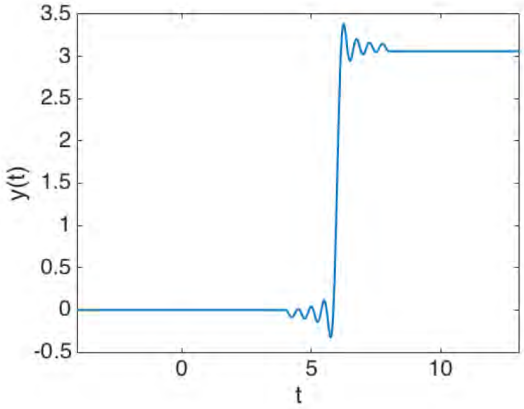
Problems 7-9 ask you to find the output of a continuous-time LTI system given the input and impulse response. The graphs below represent the choices for the output $y(t)$ in each case.



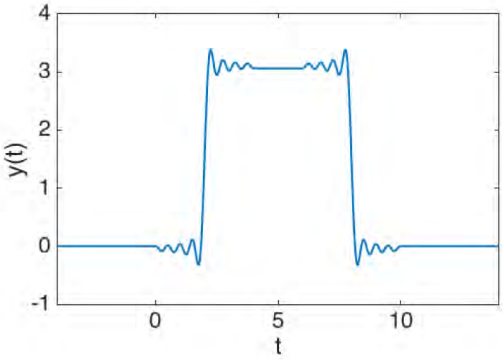
(a)



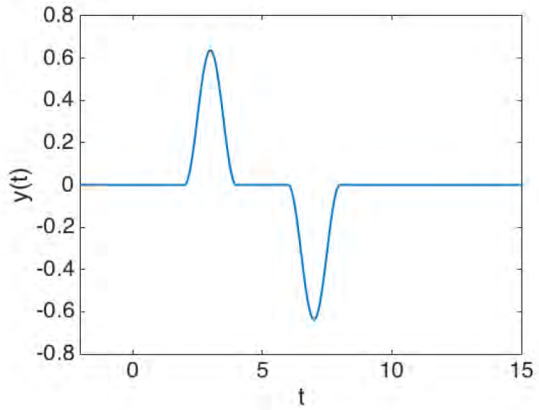
(b)



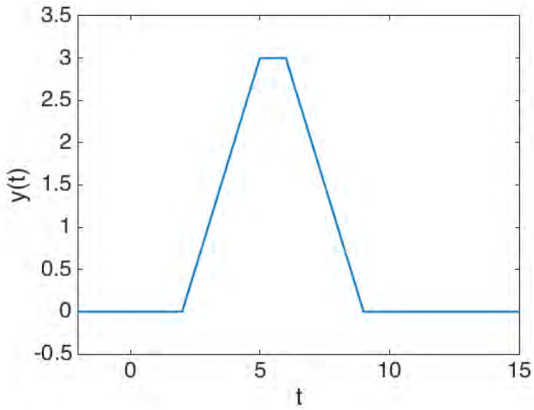
(c)



(d)

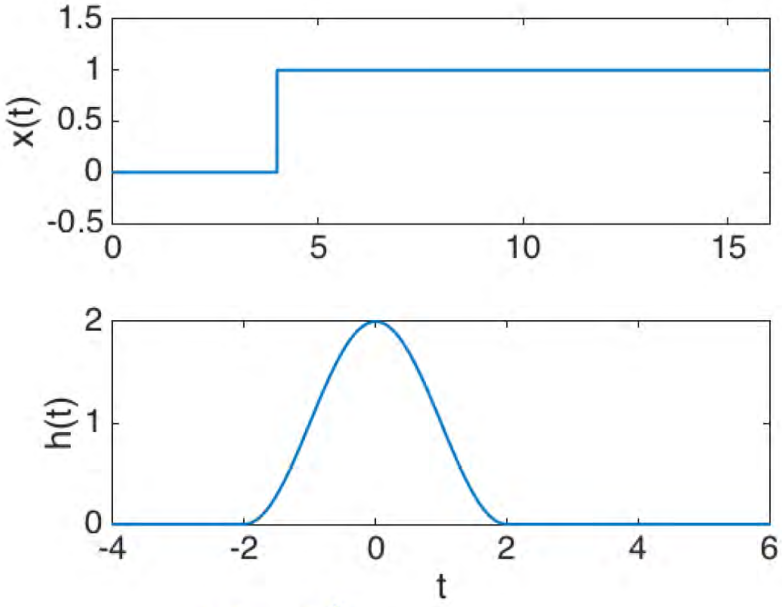


(e)

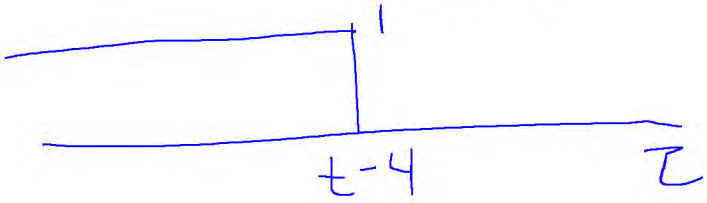


(f)

7. Identify the correct output for the input $x(t)$ and LTI system impulse response $h(t)$ depicted below.



$x(t-\tau)$

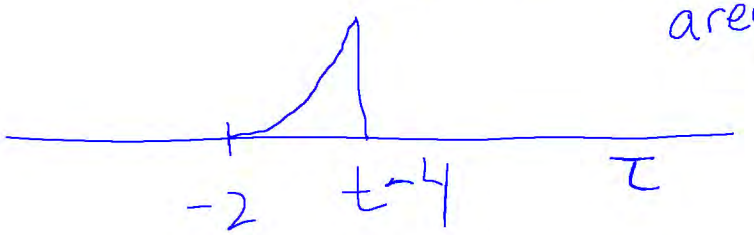


$x(t-\tau)h(\tau)=0,$

$t-4<-2$

$t<2$

$x(t-\tau)h(\tau)$

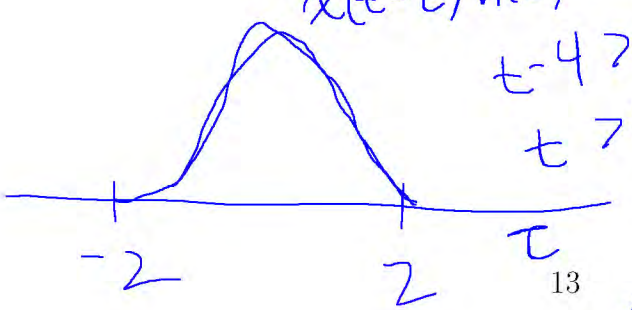


area increases monotonically as t increases until $t=6$

$x(t-\tau)\,h(\tau)$

$t-4>2$

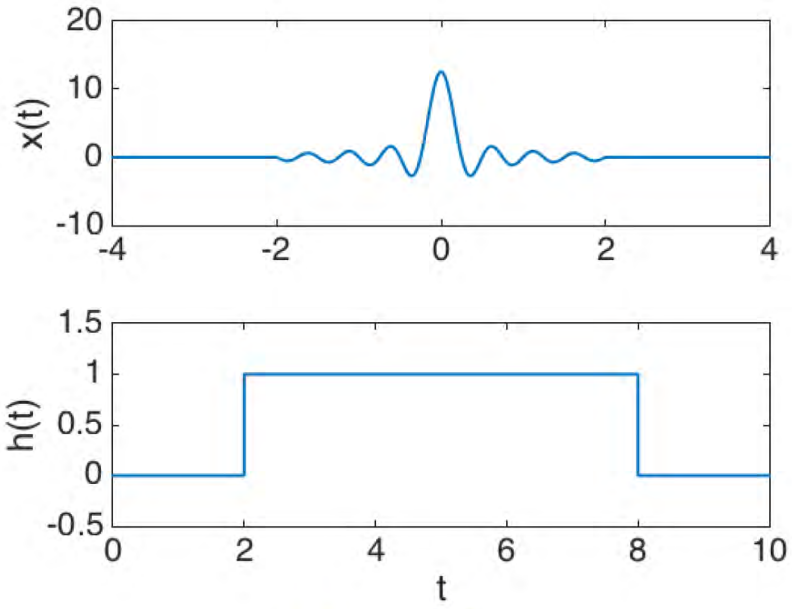
$t>6$

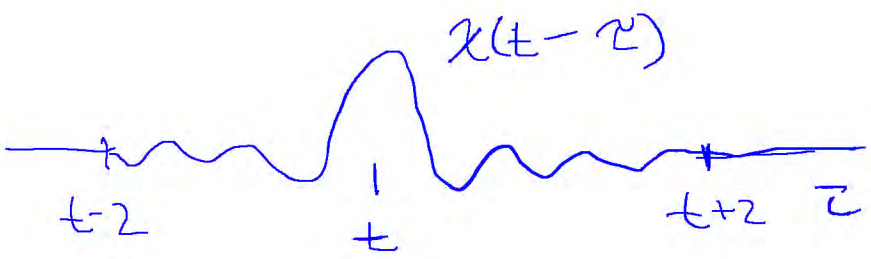


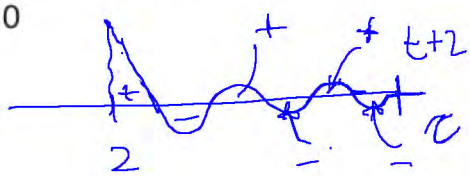
area constant as t increases for $t>6$

Case a

8. Identify the correct output for the input $x(t)$ and LTI system impulse response $h(t)$ depicted below.







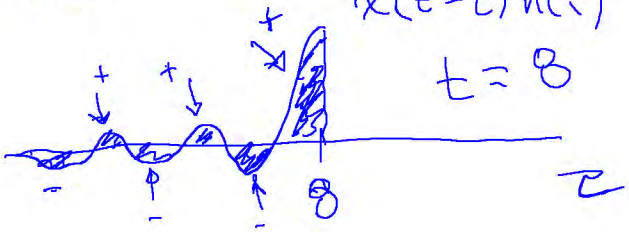
$x(t-\tau)\,h(\tau)$ zero for $t+2<2 \Rightarrow t<0 \Rightarrow y(t)=0$

and $t-2>8 \Rightarrow t>10$

area oscillates for $t>0$, then increases significantly $t\approx 2$

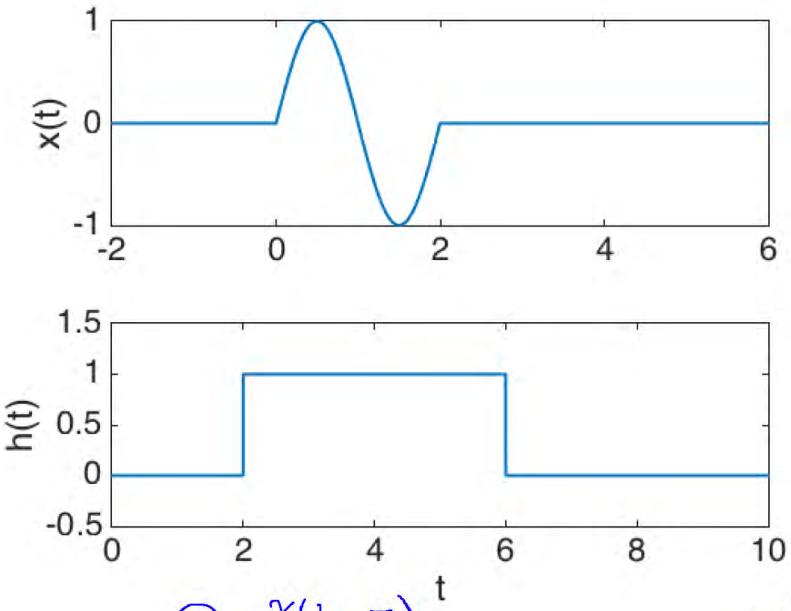
for $t>2$, area oscillates about some positive value until $t-2\simeq 2$ $(t=4)$, then stays constant until $t+2=8\,(t=6)$

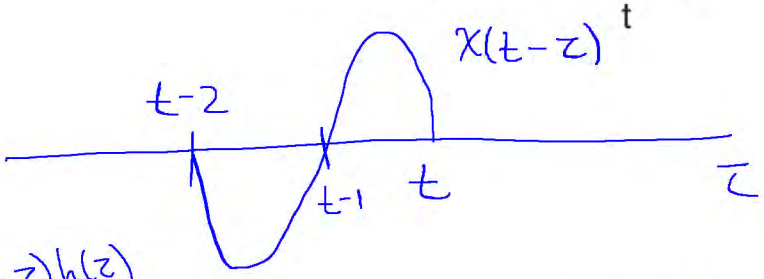
then oscillates, then decays towards zero when $t\approx 8$, and oscillates about zero until $t-2>8$



Case 5

9. Identify the correct output for the input $x(t)$ and LTI system impulse response $h(t)$ depicted below.





$x(t-\tau)h(\tau) = 0$ for:

$t < 2$

$t - 2 > 6 \quad (t > 8)$

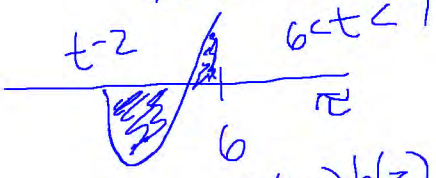
$x(t-\tau)h(\tau)$

area increases monotonically for $t > 2$, $t < 3$

area decreases to zero from $t = 3$ to $t = 4$



$x(t-\tau)h(\tau)$



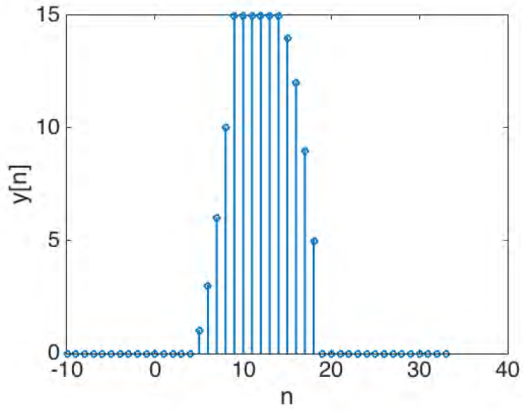
$x(t-\tau)h(\tau)$



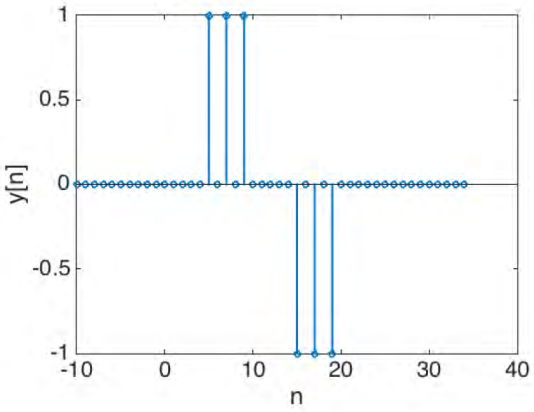
area decreases monotonically as t goes from 6 to 7, then increases to zero as t goes from 7 to 8

Case e

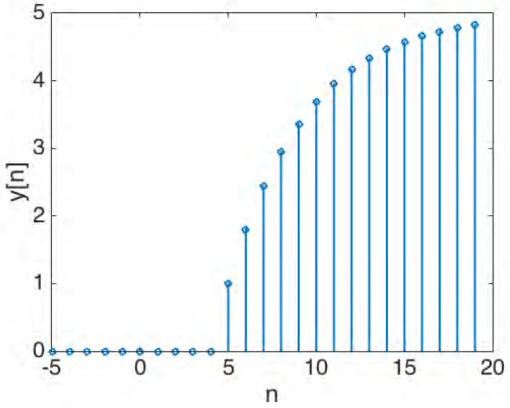
Problems 10-12 ask you to find the output of a discrete-time LTI system given the input and impulse response. The graphs below represent the choices for the output $y[n]$ in each case.



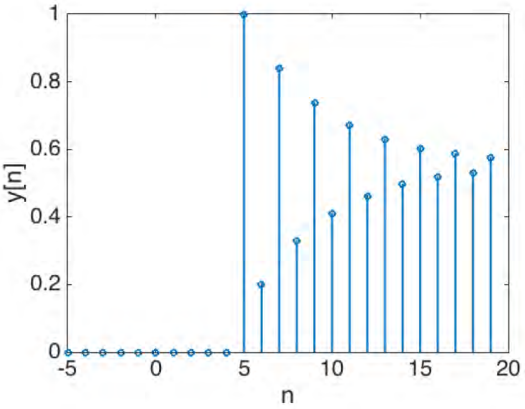
(g)



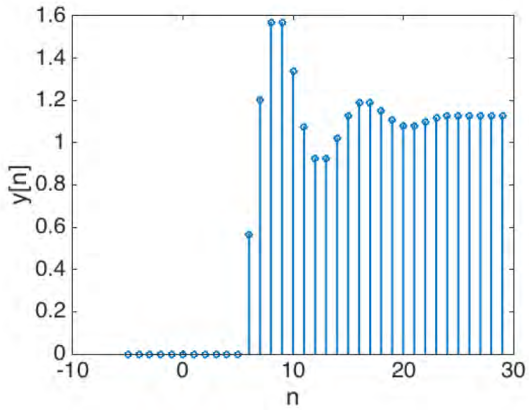
(h)



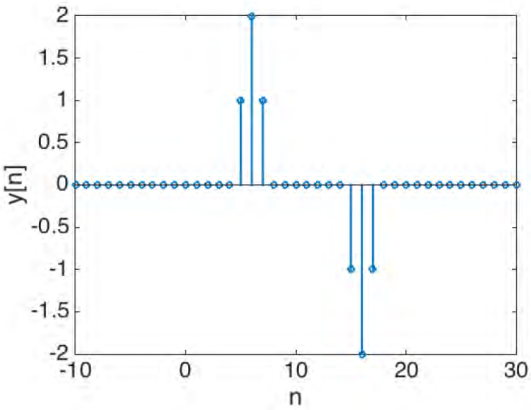
(i)



(j)

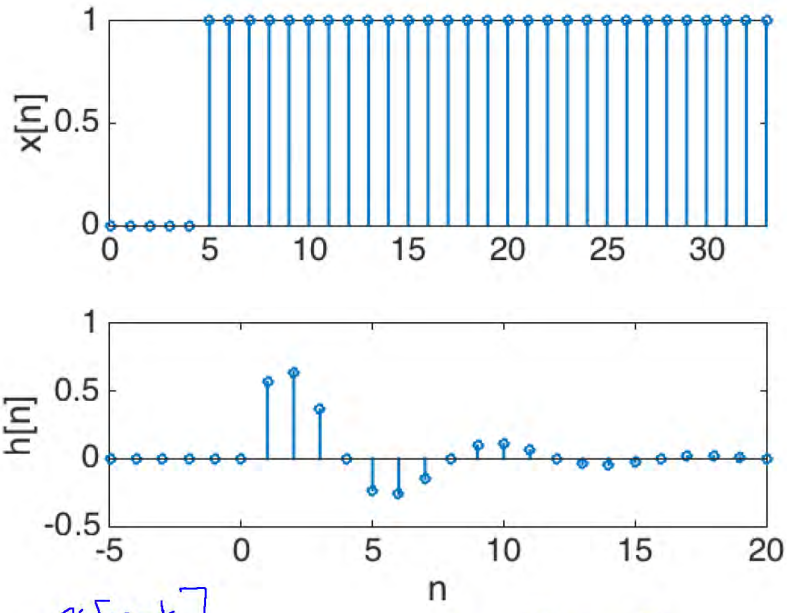


(k)

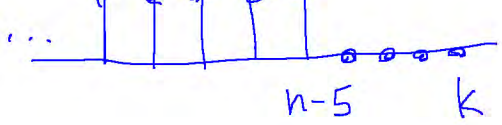


(l)

10. Identify the correct output for the input $x[n]$ and LTI system impulse response $h[n]$ depicted below.



$x[n-k]$

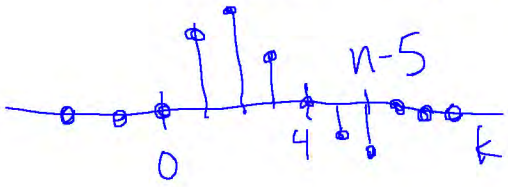


$x[n-k]h[k] = 0$ for $n-5 \leq 0$

$n \leq 5$

$\sum_{k=-\infty}^{\infty} x[n-k]h[k] = y[n]$

$x[n-k]h[k]$ $n > 5$



$y[n]$ increases til $n-5=3$

$n=8$

decreases as $n-5 \geq 5$

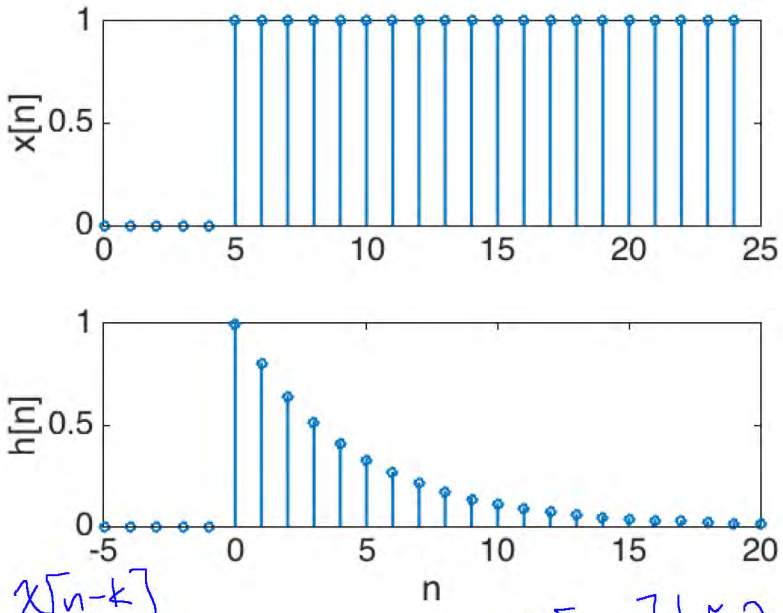
$(n \geq 10)$

then increases when $n-5=9$

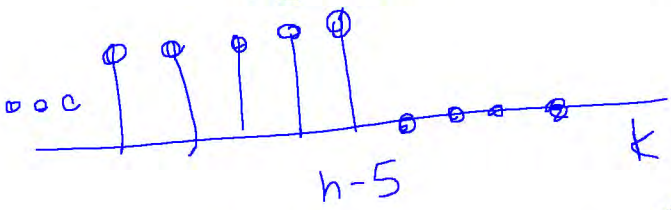
settles at a constant value, positive

Case 4

11. Identify the correct output for the input $x[n]$ and LTI system impulse response $h[n]$ depicted below.



$x[n-k]$

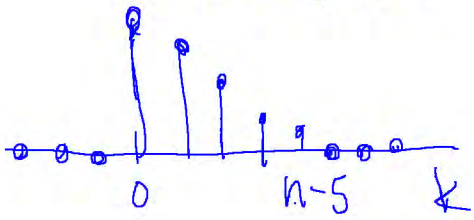


$x[n-k]h[k]=0 \quad n-5<0$

$n<5$

$y[n]=0$

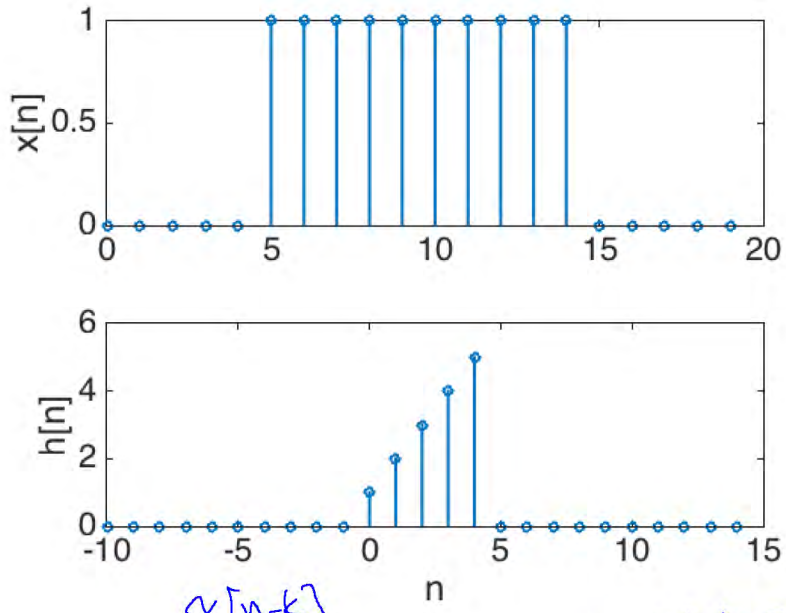
$x[n-k]h[k]$



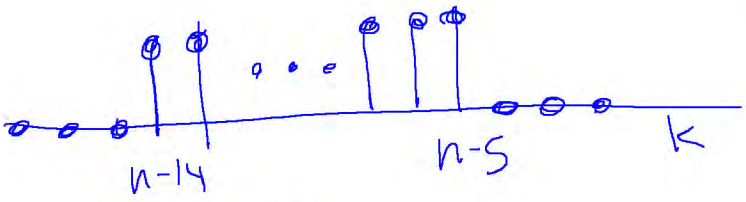
$y[n]=\sum_{k=-\infty}^{\infty} x[n-k]h[k]$ increases monotonically for $n \geq 5$

Case 1

12. Identify the correct output for the input $x[n]$ and LTI system impulse response $h[n]$ depicted below.



$x[n-k]$



$x[n-k]h[k] = 0$ for:

$n-5 < 0$ or $n < 5$

$n-14 > 4$ or $n > 18$

$$y[n] = \sum_{k=-\infty}^{\infty} x[n-k]h[k]$$

increases as $n \geq 5$ until $n-5 = 4$ $(n=9)$ then stays constant until $n-14 = 0$ where it begins to decrease